Assignment 8 Solutions

BASIC EXERCISES

9.2 $\dim(\text{Nul}(A)) + \dim(\text{Col}(A)) = 13$

~~$\dim(\text{Nul}(A)) = \#$ free variables $\leq$ dim~~

$\dim(\text{Col}(A)) = \#$ of pivot positions $\leq \min(m, n) = 10$ here

$\Rightarrow \dim(\text{Nul}(A)) = 13 - \dim(\text{Col}(A)) \geq 13 - 10 = \boxed{3}$

9.3 $\underset{4}{\text{rk } A} + \dim(\text{Nul}(A)) = 5$

$4 + \dim(\text{Nul}(A)) = 5 \Rightarrow \dim(\text{Nul}(A)) = 1$

9.4
$$\begin{bmatrix} -1 & 0 & 1 \\ 3 & 6 & 0 \\ 5 & 7 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 10 - 7 + 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 7 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\vec{v} \notin \text{Nul } A$

9.6 How many lin. ind. columns of A are there?

$$A = \begin{bmatrix} | & | & | & | \\ \vec{a}_1 & \vec{a}_2 & \vec{a}_3 & \vec{a}_4 \\ | & | & | & | \end{bmatrix} = \begin{bmatrix} 2 & 1 & -8 & 3 \\ -1 & 3 & 4 & 2 \\ 3 & 2 & -12 & 5 \\ 1 & -2 & -4 & -1 \end{bmatrix}$$

Note that $\vec{a}_3 = -4\vec{a}_1$, so we may discard one while constructing a lin. ind. set

Note also that $\vec{a}_1 + \vec{a}_2 = \vec{a}_4$.

So if we keep $\vec{a}_1$ & $\vec{a}_2$, we still capture $\vec{a}_3$ & $\vec{a}_4$ in the span.

Are $\vec{a}_1, \vec{a}_2$ lin. ind.? Yes, they are not scalar multiples.

Thus $\{\vec{a}_1, \vec{a}_2\}$ is a basis for $\text{Col}(A)$ & $\text{rk } A = 2$

9.9 $\begin{bmatrix} 0 & -2 & 2 \\ -2 & 3 & -9 \\ -1 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 7 & -7 \\ 0 & -2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

basis for Col(A): $\left\{ \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix} \right\}$

basis for Nul(A): $x_1 = -3x_3$, $x_2 = x_3$, x_3 free

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$

$\left\{ \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \right\}$

9.18 $x_1 \begin{bmatrix} 1 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} -17 \\ 15 \end{bmatrix}$

$\left[\begin{array}{cc|c} 1 & 3 & -17 \\ -5 & -1 & 15 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 3 & -17 \\ 0 & 14 & -70 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 3 & -17 \\ 0 & 1 & -5 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -5 \end{array}\right]$

$x_1 = -2$, $x_2 = -5$ $\quad [\vec{u}]_{\mathcal{B}} = \begin{bmatrix} -2 \\ -5 \end{bmatrix}$

9.20 Want a matrix that sends $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \mapsto e_1$ & $\begin{bmatrix} -3 \\ -5 \end{bmatrix} \mapsto e_2$

This is just $\begin{bmatrix} 1 & -3 \\ 2 & -5 \end{bmatrix}^{-1} = \frac{1}{-5+6} \begin{bmatrix} -5 & 3 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 3 \\ -2 & 1 \end{bmatrix}$

(In general for $\mathcal{B} \to \underset{\text{std basis}}{\mathcal{E}}$, gather basis elements into square matrix and take inverse)

↙ find $\vec{b}$ such that system infeasible

9.23
$$\left[\begin{array}{ccc|c} 1 & 1 & 5 & b_1 \\ -1 & 0 & -1 & b_2 \\ 1 & 2 & 9 & b_3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 5 & b_1 \\ 0 & 1 & 4 & b_1+b_2 \\ 0 & 1 & 4 & b_3-b_1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 5 & b_1 \\ 0 & 1 & 4 & b_1+b_2 \\ 0 & 0 & 0 & b_3-2b_1-b_2 \end{array}\right]$$

Just need anything that satisfies $b_3-2b_1-b_2 \neq 0$

Say $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $\quad 1-2-2=-3\neq 0$

TRUE/FALSE

9.24 F, $\left\{\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right\}$ & $\left\{\begin{bmatrix}-1\\0\end{bmatrix},\begin{bmatrix}0\\-1\end{bmatrix}\right\}$ are different bases for $\mathbb{R}^2$ (a subspace of itself)

9.26 F, $rk\ A \leq \min(3,6)=3 \Rightarrow \dim(Nul(A)) \geq 3$

9.28 T, $A\vec{x}=\vec{b}$ having a solution means

$x_1\vec{a}_1+\cdots+x_n\vec{a}_n=\vec{b}$ (where $\vec{a}_i$ denotes the ith column of A)

$\Leftrightarrow \vec{b} \in span(\{\vec{a}_1,\ldots,\vec{a}_n\}) = Col(A)$

9.31 F, if A implements T, $rk\ A = \underbrace{\dim(Col(A))=2}_{\text{as it maps to a plane}} \Rightarrow \dim(Nul(A))=1$

9.33 F, $A=\begin{bmatrix}0&0\\0&0\end{bmatrix}$ $\quad \dim(Nul(A))=2 > \dim(Col(A))=0$

9.35 F, $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $\quad rk\ A = 1 \quad Ax = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ only soln is $x = 0$

9.37 T, $rk\ A + \dim(Nul(A)) = n$

$\Rightarrow rk\ A^T + \dim(Nul(A^T)) = m$

these are equal

first equation minus second:

$\dim(Nul(A)) - \dim(Nul(A^T)) = n - m$

MORE DIFFICULT PROBS

9.40 Let's gather into a matrix and row-reduce to figure out linear dependencies

$$\begin{bmatrix} 3 & -3 & -1 & 8 \\ -1 & 1 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & 1 & -4 \\ 0 & 0 & 2 & -4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & -1 & 4 \\ 0 & 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$\{\vec{v}_1, \vec{v}_3\}$, $\{\vec{v}_2, \vec{v}_3\}$, $\{\vec{v}_1, \vec{v}_4\}$, $\{\vec{v}_2, \vec{v}_4\}$, $\{\vec{v}_3, \vec{v}_4\}$

These are all lin. ind. subsets of $\vec{v}_i$'s based on columns of RREF

9.43 (a) $A\vec{v} = \vec{b}$, $w \in Nul\,A \Rightarrow A\vec{w} = \vec{0}$

$A(\vec{v} + \vec{w}) = A\vec{v} + A\vec{w} = \vec{b} + \vec{0} = b$

(b) This set is not closed under scaling. Take $c \neq 0$ and assume $A\vec{x} = \vec{b}$. Then $A(c\vec{x}) = cA\vec{x} = c\vec{b} \neq \vec{b}$.

$c\vec{x} \notin \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \vec{b}\}$

(c) Sol'n set is $\{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \vec{b}\}$

If nonempty, take $\vec{o}$ such that $A\vec{o} = \vec{b}$

For U take $Nul(A)$

$\{\vec{u} + \vec{o} \mid \vec{u} \in Nul(A)\} \subseteq \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \vec{b}\}$ by (a)

$\geq$

Reverse inclusion:

for any $\vec{v}$ s.t. $A\vec{v} = \vec{b}$, $A\vec{o} = \vec{b}$ $\Rightarrow A(\vec{v} - \vec{o}) = \vec{0} \Rightarrow \vec{v} - \vec{o} \in Nul(A)$

$\vec{v} = (\vec{v} - \vec{o}) + \vec{o}$